

Seasonal Multi-Regime Machine Learning for Daily PM_{2.5} Prediction in Dhaka, Bangladesh

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Abstract—This project investigates whether seasonal specialization improves daily PM_{2.5} prediction in Dhaka, Bangladesh, compared with a single model trained across the full year. The Kaggle Dhaka air-quality dataset was processed into 1,291 usable daily observations and divided chronologically into 925 training and 366 held-out testing samples. Three calendar-based regimes were defined: Winter_Dry, Monsoon_Wet, and Transition. Random Forest, XGBoost, and LightGBM were each evaluated in two configurations: one global model and a routed multi-regime system with separate specialists for the three regimes. Global Random Forest produced the lowest absolute test error (RMSE 8.89 $\mu\text{g}/\text{m}^3$, $R^2 = 0.92$), while its multi-regime version was worse (RMSE 9.41). In contrast, specialization reduced XGBoost RMSE from 10.89 to 9.95 (8.7%) and LightGBM RMSE from 10.82 to 10.14 (6.31%). The improvement in both boosting methods was concentrated in Winter_Dry, where specialist RMSE improved by 13.17% for XGBoost and 10.65% for LightGBM. A weighted Winter XGBoost experiment increased high-pollution recall from 95.8% to 96.7% but increased RMSE from 12.83 to 13.23. The results therefore do not support universal superiority of multi-regime modelling; they instead show that the benefit of regime specialization is both algorithm- and season-dependent.

Index Terms—PM_{2.5}, air quality, Dhaka, seasonal regimes, XGBoost, Random Forest, LightGBM, time-series prediction, data mining.

I. INTRODUCTION

Fine particulate matter (PM_{2.5}) is an important air-quality indicator because sustained exposure to elevated concentrations is associated with substantial health risk [1]. Dhaka experiences strong seasonal variation in ambient particulate pollution, and the project dataset shows a visibly different PM_{2.5} distribution across dry, wet, and transition months. This raises a practical modelling question: should a machine-learning model learn the full annual distribution at once, or can it benefit from specializing in seasonal sub-distributions?

The study is designed around that question rather than around a simple model leaderboard. Three widely used tree-based algorithms are compared: Random Forest [2], XGBoost [3], and LightGBM [4]. Each algorithm is trained once as a global model and again as three seasonal specialists. The same held-out future period is then used for a paired comparison between the two configurations.

The main contribution is therefore an empirical analysis of the *effect of regime specialization*. The evidence is intentionally interpreted conservatively. XGBoost and LightGBM benefit

overall from seasonal routing, particularly in Winter_Dry, whereas Random Forest performs better globally. The project thus treats algorithm dependence as a result rather than assuming in advance that multi-regime modelling must win.

II. RELATED WORK

Recent literature has extensively demonstrated the effectiveness of tree-based machine learning algorithms, such as Random Forest and XGBoost, for predicting daily PM_{2.5} concentrations [5], [6]. However, the standard methodological approach in urban air quality prediction is to utilize a *Single Global Regime*. In studies focusing on South Asian urban centers [7] and broader regional air quality modeling, researchers typically combine all meteorological and pollutant data across all months into a single training matrix.

In these standard frameworks, seasonal variation is treated merely as an input feature (e.g., passing month or temperature as a column to the model). The underlying algorithm is forced to minimize a single unified loss function across all seasonal dynamics simultaneously. While effective for general forecasting, this single-regime approach struggles when the data distribution is highly non-stationary. As Dhaka exhibits drastic shifts in PM_{2.5} between the dry winter and wet monsoon seasons, a single global model risks underfitting the extreme high-pollution spikes of winter while overfitting the cleaner monsoon days.

Our proposed *Multi-Regime* framework addresses this gap. Instead of forcing a single model to map the entire annual distribution, we decouple the data into distinct seasonal periods, training independent models that specialize in the unique atmospheric chemistry and physical dynamics of their specific regime.

III. RESEARCH QUESTION AND HYPOTHESIS

The guiding research question is:

How does seasonal regime specialization affect daily PM_{2.5} prediction across different machine-learning algorithms, and is the benefit concentrated in particular seasonal regimes?

The refined hypothesis is that seasonal specialization can improve gradient-boosting models, especially during the high-

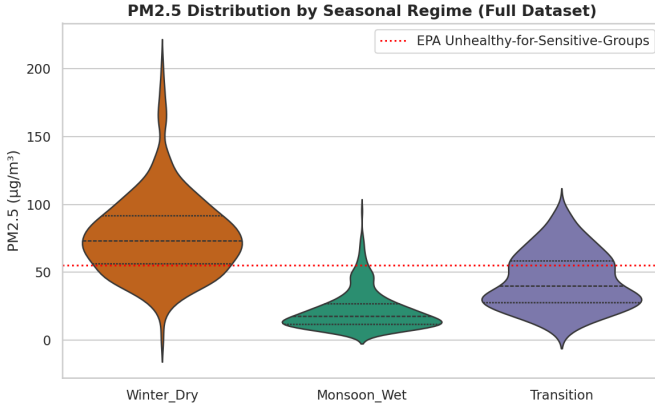


Fig. 1. PM_{2.5} distribution by the three calendar-based regimes. This notebook visualization shows the higher concentration and wider spread of Winter_Dry relative to Monsoon_Wet and Transition.

pollution Winter_Dry regime, but that the benefit may not generalize to every learning algorithm.

IV. DATASET AND EXPLORATORY MOTIVATION

The project uses the *Dhaka Air Quality Historical Dataset* hosted on Kaggle under the identifier nitirajkulkarni/dhaka-bd-1185241 [8]. The source file is `air_quality_historical.csv`. The raw dataset contains 1,298 rows. After date ordering, lag construction, and removal of rows without sufficient lag history, the modelling dataset contains 1,291 observations covering 8 August 2022 to 18 February 2026.

The regression target is daily `pm2_5`, measured in $\mu\text{g}/\text{m}^3$. The notebook uses 11 predictors: carbon monoxide, nitrogen dioxide, sulphur dioxide, ozone, aerosol optical depth, dust, UV index, day of week, PM_{2.5} lag-1, PM_{2.5} lag-7, and a three-day rolling PM_{2.5} feature.

Fig. 1 summarizes the data-level motivation for regime modelling. Winter_Dry has the highest PM_{2.5} values and visibly greater variability, Monsoon_Wet is lower, and Transition lies between them. This distributional difference motivates the experiment, but it is not itself evidence that specialist models will perform better.

V. METHODOLOGY

A. Preprocessing and Feature Construction

The date column was converted to a datetime type and sorted in ascending order. The raw data contain a small number of missing values: most air-quality variables have three missing observations and the AQI columns have four. The current notebook applies forward fill followed by backward fill before feature construction. It then generates `day_of_week`, `month`, PM_{2.5} lag-1, lag-7, and a three-day rolling mean shifted by one day.

A methodological limitation should be stated clearly. Because backward filling is currently performed before the train/test split, a later observation can theoretically influence an earlier missing entry. The chronological evaluation itself is appropriate, but a

TABLE I
CHRONOLOGICAL SPLIT BY REGIME

Regime	Training	Testing
Winter_Dry	412	152
Monsoon_Wet	298	122
Transition	215	92
Total	925	366

strict final research pipeline should perform imputation using training information only. Consequently, the present results are treated as the current experimental findings rather than as a fully leakage-audited deployment pipeline.

B. Seasonal Regime Definition

Regimes are assigned deterministically from calendar month:

- **Winter_Dry**: November–March,
- **Monsoon_Wet**: June–September,
- **Transition**: April, May, and October.

Month is used for routing but is not included as a direct prediction feature. The method is therefore a *calendar-based regime routing* approach, not a learned regime detector.

C. Chronological Train-Test Split

Random shuffling was avoided because the task is temporal prediction. The final approximately one-year period was held out for testing, using 18 February 2025 as the cutoff. This produced 925 training and 366 test observations. Time-ordered evaluation is consistent with standard forecasting practice in which future observations should not be used to evaluate models trained on later data [9].

D. Global and Multi-Regime Framework

For each algorithm, the global configuration is trained on all 925 training observations. The multi-regime configuration trains three separate models on the Winter_Dry, Monsoon_Wet, and Transition subsets. During testing, the month of each observation selects the corresponding specialist. Global and routed predictions are evaluated on exactly the same held-out rows.

To formalize the distinction between the two approaches, the **Global Model** minimizes a single loss function over the entire training dataset:

$$\text{Loss}_{\text{Global}} = \sum_{t=1}^{N_{\text{total}}} (y_t - f_{\text{global}}(X_t))^2 \quad (1)$$

where N_{total} includes data from all 12 months.

In contrast, the **Multi-Regime Framework** physically separates the training data into subsets (N_{Winter} , N_{Monsoon} , $N_{\text{Transition}}$) based on calendar routing. It minimizes separate loss functions for each regime r :

$$\text{Loss}_r = \sum_{t=1}^{N_r} (y_t - f_r(X_t))^2 \quad (2)$$

METHODOLOGY DIAGRAM

Seasonal Multi-Regime Machine Learning for PM_{2.5} Prediction in Dhaka, Bangladesh

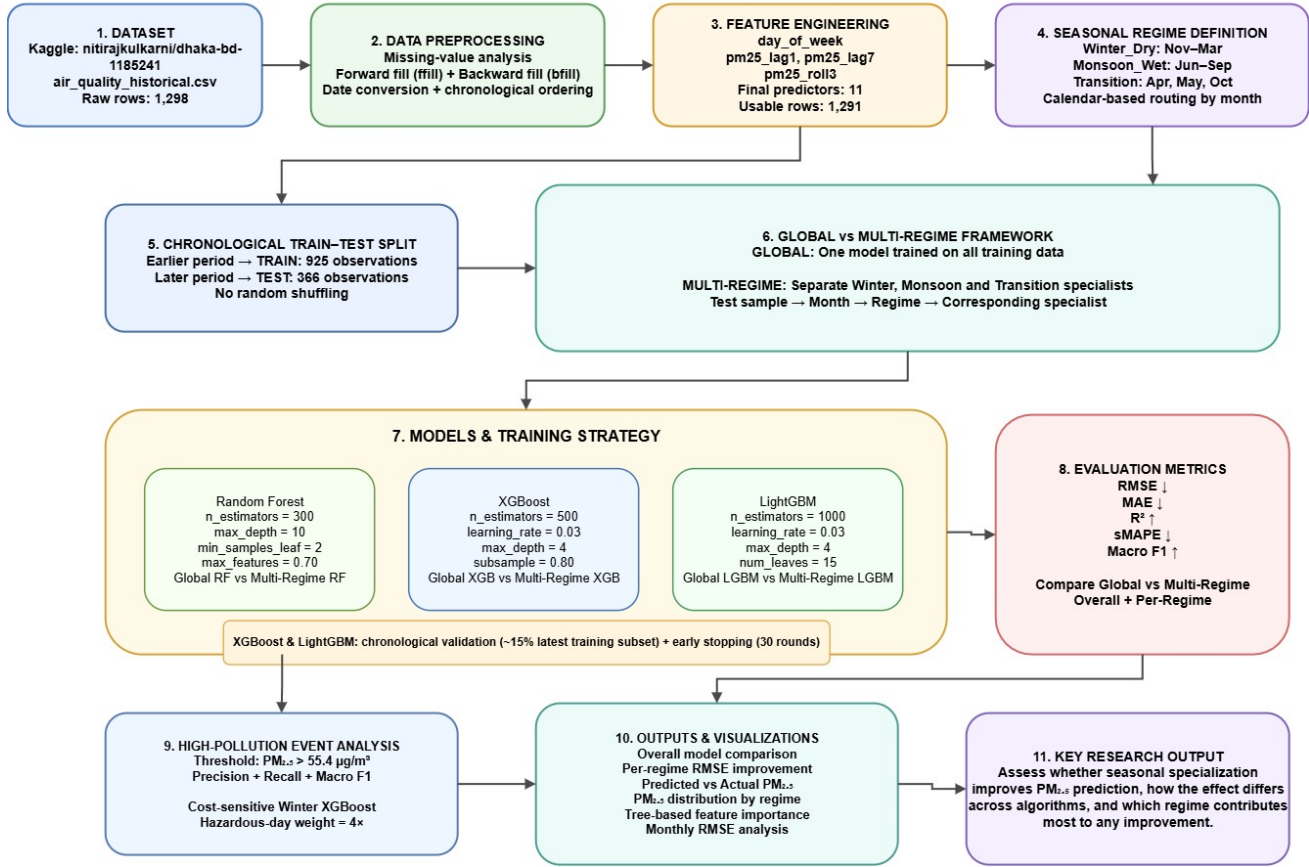


Fig. 2. Methodology used in the project. The diagram also records the current preprocessing caveat: the final strict pipeline should fit imputation on training data only.

This isolation allows the boosting trees to optimize their splits specifically for the variance present in that particular season, rather than averaging splits across conflicting seasonal patterns. During inference, observations are routed to the corresponding f_r based on their month.

Fig. 2 shows the complete project workflow, including dataset preparation, feature engineering, seasonal routing, training, evaluation, high-pollution analysis, and visual diagnostics.

E. Model Training

Random Forest uses 300 trees, a maximum depth of 10, minimum leaf size of 2, and `max_features=0.7`. XGBoost uses up to 500 estimators, learning rate 0.03, maximum depth 4, minimum child weight 5, subsampling 0.8, column sampling 0.7, L1 regularization 0.5, and L2 regularization 1.5. LightGBM uses up to 1000 estimators, learning rate 0.03, maximum depth 4, 15 leaves, minimum child samples 10, subsampling 0.8, column sampling 0.7, and the same L1/L2 regularization strengths.

For XGBoost and LightGBM, approximately the latest 15% of each training subset is reserved as chronological validation

data. Early stopping is triggered after 30 rounds without improvement in validation RMSE. Random Forest is fitted directly because it does not use sequential boosting and early stopping in the same way.

F. Evaluation Metrics

The main regression metrics are RMSE, MAE, R^2 , and sMAPE. RMSE is

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}, \quad (3)$$

and MAE is

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|. \quad (4)$$

Lower RMSE, MAE, and sMAPE indicate smaller errors; higher R^2 indicates stronger explained variation. The notebook also converts continuous PM_{2.5} predictions into AQI-style categories and reports Macro F1 as a secondary measure.

TABLE II
OVERALL HELD-OUT TEST PERFORMANCE

Algorithm	Configuration	RMSE	R^2
Random Forest	Global	8.89	0.92
Random Forest	Multi-regime	9.41	0.91
XGBoost	Global	10.89	0.88
XGBoost	Multi-regime	9.95	0.90
LightGBM	Global	10.82	0.88
LightGBM	Multi-regime	10.14	0.89

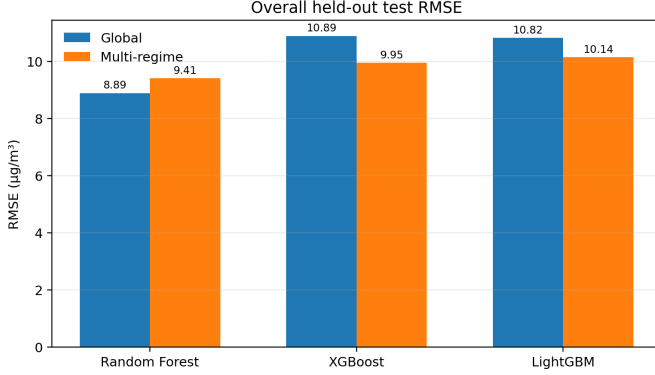


Fig. 3. Overall held-out RMSE for global and multi-regime configurations.

For regime r , the effect of specialization is reported as

$$\Delta_r = \frac{\text{RMSE}_{\text{global},r} - \text{RMSE}_{\text{specialist},r}}{\text{RMSE}_{\text{global},r}} \times 100\%. \quad (5)$$

Positive values indicate improvement from specialization; negative values indicate deterioration.

G. High-Pollution Analysis

The notebook uses a $\text{PM}_{2.5}$ threshold of $55.4 \mu\text{g}/\text{m}^3$ to create a binary high-pollution condition and computes recall and precision. These implemented cut points correspond to the AQI scheme used in the notebook and should be reconciled with the exact regulatory standard used in any future operational deployment [10]. A cost-sensitive Winter XGBoost experiment assigns four times the sample weight to observations above the threshold to test the trade-off between general regression error and sensitivity to high-pollution events.

VI. RESULTS

A. Overall Global vs Multi-Regime Performance

Table II gives the central six-model comparison. Global Random Forest is the best absolute predictor among the tested configurations with RMSE 8.89 and $R^2 = 0.92$. However, its multi-regime form is worse. XGBoost and LightGBM show the opposite pattern: both improve when routed through seasonal specialists.

The overall specialization effect is -5.9% for Random Forest, $+8.7\%$ for XGBoost, and $+6.31\%$ for LightGBM. Fig. 3 visualizes this contrast.

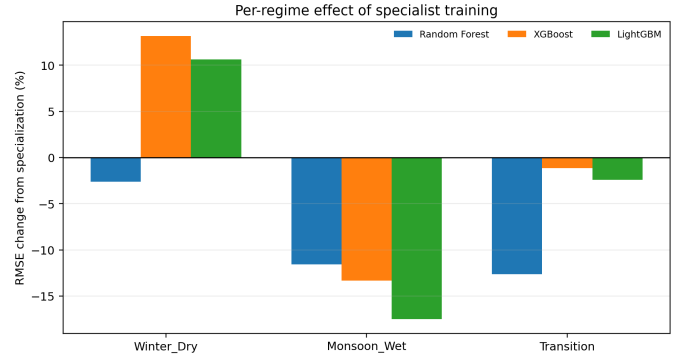


Fig. 4. RMSE change produced by specialist training in each regime. Positive values indicate improvement.

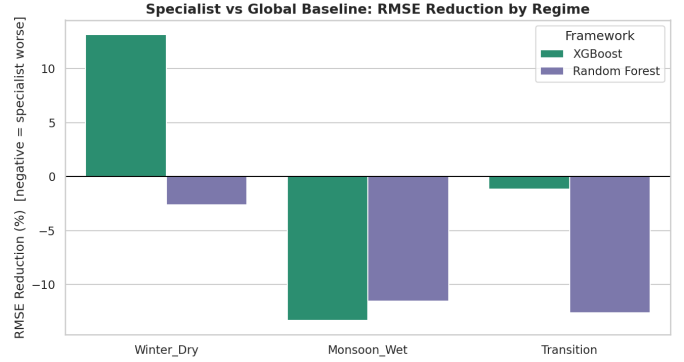


Fig. 5. Notebook-generated specialist-versus-global RMSE reduction by regime for XGBoost and Random Forest. Positive values indicate improvement; negative values indicate deterioration.

B. Per-Regime Specialization

The strongest and most consistent positive result occurs in Winter_Dry. XGBoost specialist RMSE improves from 14.78 to 12.83 (13.17%), while LightGBM improves from 14.68 to 13.12 (10.65%). Random Forest deteriorates slightly in winter and more strongly in the other two regimes. XGBoost and LightGBM also become worse in Monsoon_Wet and slightly worse in Transition.

Fig. 4 makes the algorithm dependence explicit. The positive Winter bars for both boosting methods are the clearest evidence in favour of regime specialization; the negative Random Forest bars show why a universal claim would be incorrect.

C. Additional Metrics

The notebook computes MAE, sMAPE, and Macro F1 for XGBoost and Random Forest. Multi-regime XGBoost improves RMSE, MAE, and R^2 , but sMAPE and Macro F1 are slightly worse. Random Forest is worse after specialization on all reported metrics.

D. Prediction Diagnostics

Fig. 6 shows predicted versus actual $\text{PM}_{2.5}$ for the multi-regime XGBoost and Random Forest systems. Predictions generally follow the observed range, but high observed $\text{PM}_{2.5}$

TABLE III
PER-REGIME GLOBAL VERSUS SPECIALIST RMSE

Algorithm	Regime	Global RMSE	Specialist RMSE	Change (%)
Random Forest	Winter_Dry	11.29	11.58	-2.61
Random Forest	Monsoon_Wet	4.80	5.36	-11.56
Random Forest	Transition	8.55	9.63	-12.64
XGBoost	Winter_Dry	14.78	12.83	+13.17
XGBoost	Monsoon_Wet	4.75	5.39	-13.32
XGBoost	Transition	9.01	9.12	-1.16
LightGBM	Winter_Dry	14.68	13.12	+10.65
LightGBM	Monsoon_Wet	4.67	5.49	-17.50
LightGBM	Transition	8.98	9.19	-2.40

TABLE IV
ADDITIONAL METRICS AVAILABLE FROM THE NOTEBOOK

Configuration	RMSE	MAE	R^2	sMAPE	Macro F1
XGB Global	10.89	7.56	0.88	15.99	0.59
XGB Multi	9.95	7.18	0.90	16.25	0.57
RF Global	8.89	6.50	0.92	15.00	0.58
RF Multi	9.41	6.88	0.91	16.20	0.56

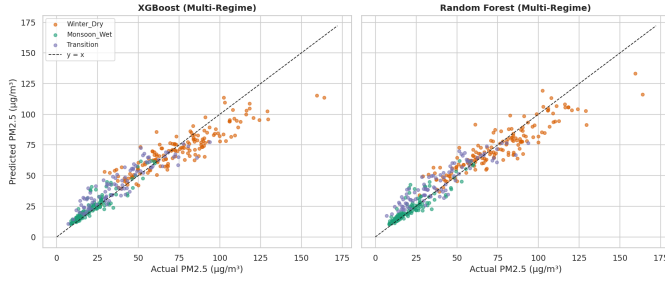


Fig. 6. Notebook-generated predicted-versus-actual $PM_{2.5}$ plots for the multi-regime XGBoost and Random Forest systems, coloured by seasonal regime. The diagonal line represents perfect agreement.

values tend to be compressed toward lower predicted values. This suggests underprediction of extreme events, a limitation that matters even when average regression metrics are strong.

E. Winter Time-Series Diagnostic

The notebook also plots the first 150 held-out Winter_Dry observations to compare the actual $PM_{2.5}$ trajectory with the global XGBoost and Winter specialist predictions. As shown in Fig. 7, both models broadly follow the temporal pattern, while the largest pollution spikes remain difficult to reproduce exactly. This plot is useful as a temporal diagnostic because it shows where numerical summary metrics may hide short periods of larger underprediction or overprediction.

F. Tree-Based Feature Importance

Fig. 8 reports the feature-importance values generated in the notebook for regime-specific XGBoost and Random Forest models. Lagged $PM_{2.5}$ and pollutant-related variables receive substantial importance in several regimes, although the ordering differs by algorithm and season. These values are descriptive

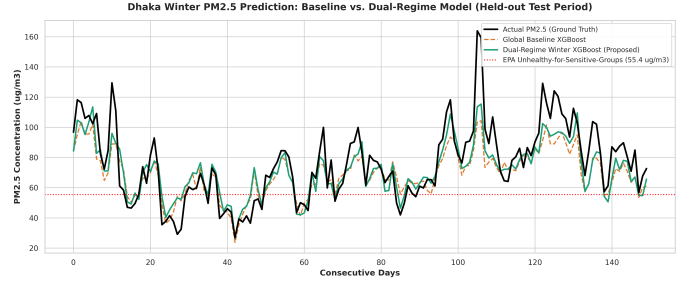


Fig. 7. Notebook-generated Winter_Dry time-series comparison for the first 150 held-out observations: actual $PM_{2.5}$, global XGBoost, and Winter specialist predictions.

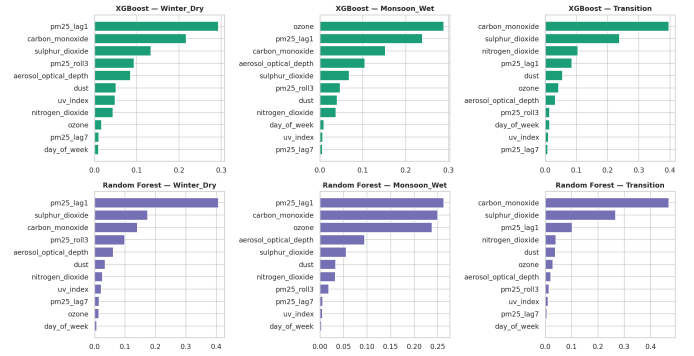


Fig. 8. Notebook-generated tree-based feature importance for regime-specific XGBoost and Random Forest models.

model importances and should not be interpreted as causal effects.

G. Monthly Error Profile

The notebook calculates monthly RMSE for the global and multi-regime XGBoost and Random Forest systems. Fig. 9 shows that error varies substantially by month, with larger values in the early-year winter period and lower values during much of the monsoon period. Because this is a single held-out year, the monthly profile is treated as a diagnostic rather than as evidence of a stable long-term monthly pattern.

H. AQI-Category Diagnostic

Continuous $PM_{2.5}$ predictions were also converted to AQI-style categories in the notebook. Fig. 10 compares confusion

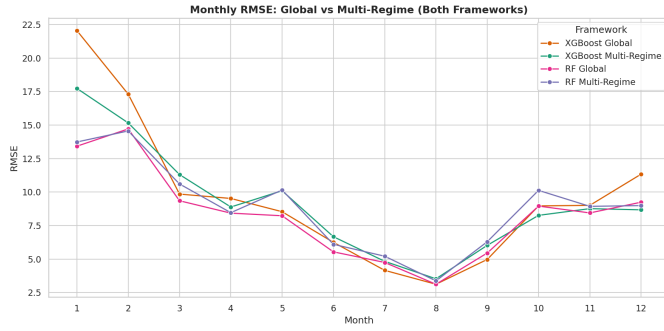


Fig. 9. Notebook-generated monthly RMSE for global and multi-regime XGBoost and Random Forest models over the held-out test period.

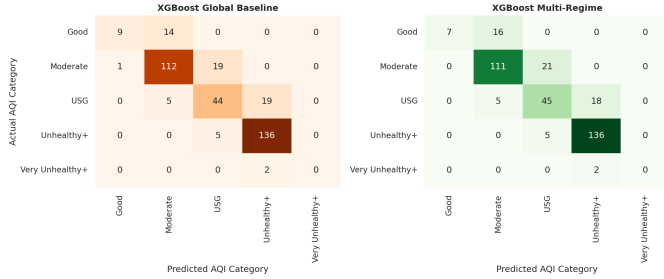


Fig. 10. Notebook-generated AQI-category confusion matrices for global and multi-regime XGBoost predictions.

matrices for the XGBoost global baseline and multi-regime configuration. This analysis is secondary to the regression task, but it helps show where numerical errors translate into category-level misclassification.

I. High-Pollution Trade-off

For Winter XGBoost, the unweighted specialist obtains RMSE 12.83 and high-pollution recall 95.8%. Giving four times the weight to high-pollution training samples increases recall to 96.7% but worsens RMSE to 13.23. The weighted model therefore improves sensitivity to the selected extreme-event threshold at the cost of continuous prediction accuracy.

VII. DISCUSSION

The results answer the research question in a more nuanced way than the original hypothesis. Seasonal specialization is not universally beneficial. Global Random Forest remains the strongest absolute model and every Random Forest specialist performs worse than its global counterpart. In contrast, XGBoost and LightGBM both improve overall under regime routing, and in both cases the gain is driven primarily by Winter_Dry.

This consistency across two independent boosting implementations is the most compelling support for the regime-aware idea. Winter is also the regime with the highest observed $PM_{2.5}$ distribution and the largest variability, so the EDA and modelling results point in the same direction. However, the present experiments establish an association between seasonal

TABLE V
COST-SENSITIVE WINTER XGBOOST TRADE-OFF

Winter model	RMSE	Recall
Unweighted specialist	12.83	95.8%
4× weighted specialist	13.23	96.7%

specialization and predictive performance; they do not prove a causal atmospheric mechanism.

The results also show why multiple metrics are necessary. XGBoost multi-regime modelling improves RMSE, MAE, and R^2 , yet does not improve sMAPE or Macro F1. Similarly, the cost-sensitive Winter model improves recall but worsens RMSE. A model should therefore be described as better only with respect to a defined objective, not as universally superior.

VIII. LIMITATIONS AND FUTURE WORK

Several limitations remain. First, the current imputation order should be made strictly leakage-safe by fitting all preprocessing using training information only. Second, the study relies on one final held-out period. Rolling or expanding-window validation is needed to determine whether the Winter specialization effect repeats across different future windows. Third, a persistence baseline, $\hat{y}_t = y_{t-1}$, should be added because lag-1 $PM_{2.5}$ is already a strong predictor. Fourth, paired error analysis and bootstrap confidence intervals would help quantify whether the reported RMSE and MAE differences are statistically robust. Fifth, the current feature-importance plots are tree-based importances rather than causal explanations; SHAP analysis would provide stronger model-level interpretability. Finally, the present regime labels are fixed by month. Future work could compare this design with data-driven or meteorology-aware regime detection.

IX. CONCLUSION

This study evaluated seasonal multi-regime modelling for daily $PM_{2.5}$ prediction in Dhaka using Random Forest, XGBoost, and LightGBM. The global Random Forest achieved the best absolute performance, while specialization hurt Random Forest. XGBoost and LightGBM, however, both benefited overall from calendar-based regime routing, with their clearest gains in Winter_Dry. The evidence therefore supports a specific conclusion: regime-aware modelling can benefit gradient-boosting approaches in this dataset, particularly during Dhaka's high-pollution winter period, but the effect is strongly algorithm-dependent and should not be generalized to all models or seasons.

DATA AND CODE AVAILABILITY

The dataset is publicly available from Kaggle [8]. Project code is maintained in a private GitHub repository for course submission: [Private repository URL]. The LaTeX source, the complete set of figures generated in the submitted notebook, the editable methodology diagram, and the Zotero-compatible BibTeX database are included with the submission package.

REPRODUCIBILITY NOTE

The numerical results in this report are taken from the supplied project notebook `final_file(3).ipynb`. Before final archival submission, the notebook should be re-run end-to-end after moving imputation inside a strictly chronological preprocessing pipeline, and the resulting metrics should be checked for any changes.

GENERATIVE AI USAGE STATEMENT

During the preparation of this work, the authors utilized Generative AI tools to assist with document formatting, text refinement, and code debugging. The core experimental design, regime routing logic, and data analysis are the original work of the authors. AI was utilized to draft the IEEEtran L^AT_EX source code, resolve double-column formatting overlaps, formalize the loss function equations (Eq. 1 and 2), and refine the academic prose based on the authors' Jupyter Notebook outputs. Python code generation via AI was limited to syntax assistance and plotting boilerplate. The authors reviewed and edited all AI-generated content and take full responsibility for the integrity and accuracy of the final submission.

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